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Summary of problem

As a new graduate, starting at age 25, with a starting salary of $66,000, I plan to retire at 65 and to secure an annual retirement income of 70% of my final salary. This plan consists of a 4% annual salary growth, and a 7% investment return during working years that turn into 3.5% in retirement. Inflation is set at 2.5% annually and I plan to live for 30 years in retirement. I withdrawal the funds needed at the beginning of each year and taxes will be excluded for this model.

Methods and Model

Key information used:

- Initial Salary: $66,000
- Salary Growth Rate: 4% annually
- Working Years: 40 years (25 to 65)
- Retirement Duration: 30 years
- Investment Return (Working Years): 7% annually
- Investment Return (Retirement): 3.5% annually
- Inflation Rate: 2.5% annually
- Retirement Income Goal: 70% of final salary
- Withdrawal Timing: Beginning of each year
- Taxes: Ignored

Key equations used:

- Final Salary = Initial Salary * $(1+0.04)^{40}$
- Annual Retirement = Final Salary * 0.7
- PV_Retirement Withdrawals = $Annual\ Retirement\ Income * \left[\frac{1-(1+0.35)^{-30}}{0.35}\right] * (1+0.35)$
- Annual Savings Needed = $\frac{PV*0.07}{(1+0.07)^{40}-1}$
- Savings Percentage = Annual Savings Needed / Initial Salary

Questions

1. What fraction of your annual salary as a percentage do you need to save to meet your retirement expenses?
    a. About 32%
2. Plot and comment on the annual income ($/yr), annual savings ($/yr), and retirement account balance ($) as a function of time over your 40-year career.

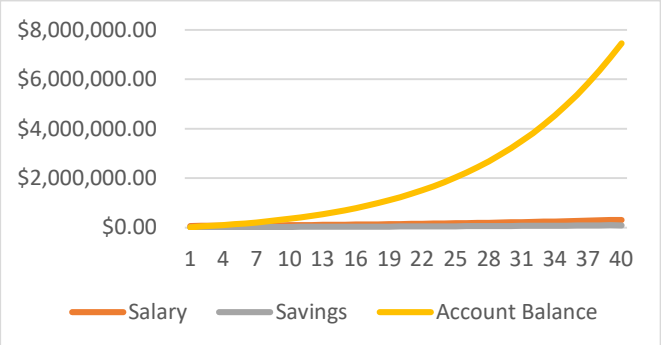


Figure 1: Plot of annual income, annual savings, and retirement account over 40 years

a. The annual income grows steadily due to the 4% salary growth and the annual savings also increase as the salary increases. The retirement account balance grows significantly

over the 40-year accumulation phase, due to the combined effect of salary growth and investment returns.

3. Plot and comment on the annual withdrawals ($/yr) and retirement account balance ($) as a function of time over your 30-year retirement.

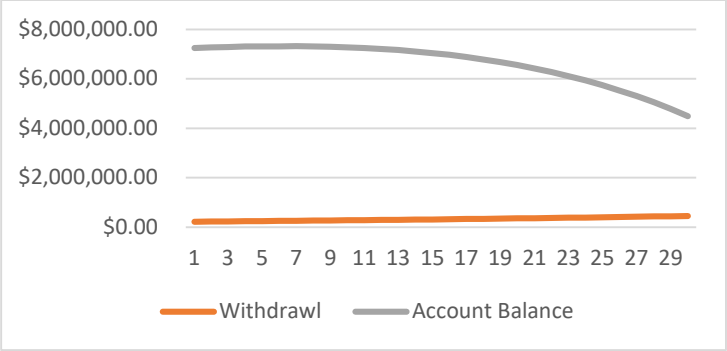


Figure 2: Plot of annual withdrawals and retirement balance over 30 years

   a. During retirement years, the account balance decreases each year as withdrawals are made to cover the retirement income needs. The withdrawals are adjusted for inflation (2.5%), leading to increasing withdrawals over time. The balance decreases due to the combination of withdrawals and investment returns (3.5%) during retirement.

4. Perform a sensitivity analysis to study the impacts of salary growth, investment growth, and inflation rate (separately) on the quality of your retirement.

Salary Growth %	Final Salary at 65	Required Annual Savings	Final Account Balance
2%	$242,024.65	$23,698.32	$5,572,129.47
4%	$316,867.36	$21,149.93	$7,459,140.61
6%	$410,298.63	$18,781.14	$9,290,679.45

Investment Return (%)	Final Account Balance at 65	Required Annual Savings
5%	$6,205,024.85	$24,227.10
7%	$7,459,140.61	$21,149.93
9%	$9,097,655.96	$18,251.15

Inflation Rate (%)	Final Account Balance at 65	Annual Withdrawals in Year 1	Remaining Balance at End of 30 Years
2%	$7,459,140.61	$221,807.15	$4,497,136.35
2.5%	$7,459,140.61	$227,352.33	$4,145,740.31
3%	$7,459,140.61	$233,036.14	$3,791,212.89

Table 1: Tables of my sensitivity analysis

Executive Summary

With my starting salary of $66,000 at 25, if I save about 32% of my salary each year, I should be able to comfortably retire by the age of 65. This is based on calculating 32% of my annual salary to meet my target retirement income, which will be 70% of my final salary at 65. This is assuming my salary grows 4% every year, I had a 7% investment return during the accumulation phase, and 2.5% inflation during retirement. The sensitivity analysis highlights the significant impact of investment returns and salary growth on retirement outcomes